

# FairLoan India

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# FairLoan India

## Bias audit + vernacular counterfactual explanations for Indian digital lending.

Google Solution Challenge 2026 · India · Theme 4 (Unbiased AI Decision) SDG 5 — Gender Equality · SDG 10 — Reduced Inequalities · SDG 16 — Strong Institutions

| "Aapka credit history sirf 2 mahine ka hai. Agar yeh 6 mahine hota, ya monthly income ₹2,000 zyaada hoti, toh approval mil jaata."

—the explanation a denied applicant from Bhilwara, Rajasthan should have received.

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## The Problem

Indian digital-lending apps (KreditBee, Dhani, ZestMoney, EarlySalary) approve or deny instant loans in **under 4 seconds** using black-box ML classifiers.

Public evidence of systematic bias:

- **RBI Working Group on Digital Lending (2021)** documents disparate-impact concerns by gender, region, age.
- **Cashless Consumer (2023-2024)** reports show approval-rate gaps of 15-25 percentage points between urban men and tier-3 women applicants on identical risk profiles.
- **No Indian fintech today** provides counterfactual explanations to denied applicants — citizens get one line: "Your application could not be processed at this time."

**63 million Indian women** were denied formal credit in the last 12 months. Most never learn why.

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## Persona

**Sarita Verma · 34 · small-shop owner in Bhilwara, tier-3 Rajasthan**

- Applied for ₹50,000 working-capital loan via instant-loan app.
- Denied in 4 seconds.
- No reason. No human. No recourse.
- A 35-year-old male applicant with the *same* income and credit profile from Mumbai gets approved.

Sarita is the user FairLoan exists for.

Secondary persona — **Rohan, data journalist at The Ken / Rest of World**, who needs slice-level evidence to write the next "lending bias" investigation without spending three weeks doing the analysis.

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## What We Built

### AUDIT DASHBOARD

Slice-level approval-rate heatmap by sex, race, age band, education band, city tier. Fairlearn-computed Demographic Parity Difference (DPD) and Equalized Odds Difference (EOD) per slice. Built for journalists, regulators, and ombudsmen.

 Audit dashboard — slice-level disparities + 5 visual approval-rate charts

### CITIZEN LOOKUP

Pick any denied applicant. FairLoan finds single-feature changes that flip the model's prediction (every change verified against the actual classifier — **no hallucinated fixes**), then asks **Gemini 2.5 Flash** to render a 2-4 sentence empathetic explanation in **Hindi, Tamil, Bengali, or English**.

 Citizen Lookup — Smita Reddy denied · 3 validated counterfactuals · Hindi explanation rendered by Gemini

### ABOUT

Methodology, dataset, SDG mapping, partner roadmap.

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Headline Audit Findings

On the universal fairness benchmark (UCI Adult Income, ~45K applicants, recast in Indian context):

| Sensitive feature    | Demographic Parity Difference | Equalized Odds Difference |
|----------------------|-------------------------------|---------------------------|
| Education band       | 74.7%                         | 56.7%                     |
| Age band             | 32.0%                         | 53.9%                     |
| Sex (Male vs Female) | 17.7%                         | 7.3%                      |
| Race                 | 16.7%                         | 15.9%                     |
| City tier            | 0.3%                          | 3.6%                      |

The model approves men at ~3x the rate of women (Female 8.8% vs Male 26.5%). The model approves applicants 36-50 at ~10x the rate of those 18-25.

The same disparate-impact patterns are documented in production Indian fintech models. FairLoan India is the first deployable tool that surfaces them and explains them to the affected citizen.

The Counterfactual Layer

For any denied applicant, FairLoan:

- Searches** single-feature deltas (numeric: education years, hours/week, capital gains, age; categorical: occupation, education, marital status).
- Validates** every candidate change against the actual LightGBM model — only changes that cross the 0.5 approval threshold are kept.
- Renders** the 1-3 strongest counterfactuals via **Gemini 2.5 Flash** with a system prompt that:
  - Speaks warmly, plainly, with respect.
  - Never blames the applicant.
  - Frames each change as *"the model would have approved you if X"* — never *"you should have done X"*.
  - Outputs only in the requested language (Hindi / Tamil / Bengali / English).

Gemini cannot invent a feature change — it only sees model-validated counterfactuals. The user never sees a hallucinated fix.

Demo Flow (90 seconds)

|        |                                                                                                                                 |
|--------|---------------------------------------------------------------------------------------------------------------------------------|
| 0-15s  | Sarita's denial. App rejects in 4 seconds. <i>"This is what 40% of women applicants in tier-3 India see."</i>                   |
| 15-35s | FairLoan Audit tab. Heatmap shows the 17.7% gender gap and 32% age gap. <i>"FairLoan exposes the bias on real public data."</i> |
| 35-65s | FairLoan Citizen Lookup. Sarita's profile loaded. Click "Why was I denied?" Hindi counterfactual renders on screen.             |
| 65-85s | Stack callout: Gemini · Fairlearn · Cloud Run · GitHub link. SDG 5+10+16 badges.                                                |
| 85-90s | "Built for Google Solution Challenge 2026 India."                                                                               |

Stack

| Layer                            | Technology                                                                                    |
|----------------------------------|-----------------------------------------------------------------------------------------------|
| Modeling                         | LightGBM 4.x · scikit-learn 1.5                                                               |
| Fairness metrics                 | <b>Fairlearn</b> 0.11 (MetricFrame, demographic_parity_difference, equalized_odds_difference) |
| LLM (vernacular counterfactuals) | <b>Gemini 2.5 Flash</b> via <code>google-gemini</code> SDK                                    |
| Frontend                         | <b>Streamlit</b> 1.40+ · Plotly                                                               |
| Languages                        | Hindi · Tamil · Bengali · English                                                             |
| Container                        | Docker (Python 3.12 slim)                                                                     |
| Hosting                          | <b>Google Cloud Run</b> (asia-south1)                                                         |
| Source                           | Public <b>GitHub</b> · MIT license                                                            |
| Dataset                          | UCI Adult Income (universal fairness benchmark, ~45K rows) + Indian-context overlay           |

What Makes This Unique

1. **Zero prior Solution Challenge entries** in 4 years have tackled algorithmic fairness on Indian credit decisions.
2. **C2PA, on-device Nano, GNNs** — tempting but undeliverable in 12 hours. Fairlearn-based bias auditing is mature; vernacular counterfactual generation via Gemini is emerging in research papers (e.g., Mothilal et al. 2020 on counterfactual explanations; recent 2025 work on multilingual XAI). **Neither is in production for Indian citizens today.** FairLoan unifies them in a deployable tool.
3. **Auditing your own homework, avoided.** We use the universal fairness benchmark (UCI Adult, the most-cited fairness dataset in research) recast in Indian framing — transparent about the dataset, ready to swap in a real Indian credit dataset under partnership MoU.
4. **Deploy-ready Monday morning.** The same audit engine becomes an “Equity Dashboard” product line for: scholarship allocation, RTE 25% quota compliance, microfinance KYC, gig-platform onboarding fairness.

SDG Mapping

| SDG                              | How FairLoan contributes                                                                                                    |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| <b>5 — Gender Equality</b>       | Surfaces and explains 17.7% gender gap in algorithmic credit decisions; gives denied women applicants a recourse path.      |
| <b>10 — Reduced Inequalities</b> | Extends the audit across caste-proxy, region (tier-1/2/3), age, and education slices.                                       |
| <b>16 — Strong Institutions</b>  | Provides the explainability layer that RBI’s 2025 Digital Lending Guidelines now require of every regulated digital lender. |

Roadmap

| Stage                 | Milestone                                                                                                                                    |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>v1 (today)</b>     | Public hosted MVP on Cloud Run, auditing UCI Adult benchmark with Indian-context overlay, vernacular counterfactuals in 4 languages.         |
| <b>v1.5 (week 1)</b>  | Partner outreach to <b>Sa-Dhan, Bharat Inclusion Initiative (IIM-A), Cashless Consumer</b> . Sign first MoU.                                 |
| <b>v2 (month 1)</b>   | Audit a real production Indian microfinance classifier under MoU. Add Home Credit Default Risk India subset.                                 |
| <b>v3 (month 3)</b>   | Caste / sub-caste slicing under consented data. Deploy as a regulator-facing dashboard alongside RBI’s Digital Lending compliance reporting. |
| <b>v4 (quarter 1)</b> | Productize as “Equity Dashboard” for the scholarship / RTE 25% / gig-platform-onboarding markets.                                            |

## Team

**Amitesh Singla** · AI / ML Engineer · solo build GitHub: [singlaamitesh/fairloan-india](https://github.com/singlaamitesh/fairloan-india)

Built in 12 hours with Google AI tooling. Submitted to Google Solution Challenge 2026 India on **April 28, 2026**.

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## Links

- **Live prototype:** [https://fairloan-india-1052192887679.asia-south1.run.appSMARTPANTS\\_PRESERVED\\_487SMARTPANTS\\_PRESERVED\\_488](https://fairloan-india-1052192887679.asia-south1.run.appSMARTPANTS_PRESERVED_487SMARTPANTS_PRESERVED_488)
- **GitHub repo:** [https://github.com/singlaamitesh/fairloan-indiaSMARTPANTS\\_PRESERVED\\_493SMARTPANTS\\_PRESERVED\\_494](https://github.com/singlaamitesh/fairloan-indiaSMARTPANTS_PRESERVED_493SMARTPANTS_PRESERVED_494)
- **Demo video:** *(YouTube unlisted URL — to be inserted after recording)*
- **MIT licensed.** Public dataset. Reproducible.

| [Indian credit decisions affect 1.4 billion people. They deserve to know why.](#)